

Implementation

Describe the program line by line via comments.

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/*
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  Menu-Driven Program to Display ASCII Values
  Directions: Create a menu-driven program that allows the user to select
  between displaying uppercase letters,
  lowercase letters, or exiting the program
  The program must be able to:
  - Display the following menu:
    - 1. Uppercase Letter (Call DisplayUppercaseASCII() to display
      uppercase letters (A-Z) with ASCII values
    - 2. Lowercase Letter (Call DisplayLowercaseASCII() to display
      lowercase letters (a-z) with ASCII values
    - 3. EXIT PROGRAM (Terminate the program)
  - The user must select an option from the menu
  - The program must continue until the user selects the EXIT option
  - Each function must display one letter and its ASCII value per line

  Sample Output:
  A = 65
  B = 66
  ...
  Z = 90
*/

// ===== HEADER =====
#include <iostream> // Provides input/output streams (cin, cout)
using namespace std; // Bring standard names into scope (e.g., cout vs std::cout)

// ===== FUNCTION PROTOTYPES =====
void DisplayUppercaseASCII(); // Display uppercase ASCII values and characters
void DisplayLowercaseASCII(); // Display lowercase ASCII values and characters

// ===== MAIN =====
// 1. Program execution starts here
int main() {
  // 1.1 Declare and initialize choice
  int choice = 0;

  // 1.2 Loop until user chooses EXIT
  while (choice != 3) {
    // 1.3 Display menu
    cout << "\n===== ASCII MENU =====\n";
  }
}

```

```
cout << "1. Uppercase Letters (A-Z)\n";
cout << "2. Lowercase Letters (a-z)\n";
cout << "3. Exit Program\n";

// 1.4 Get user input
cout << "\nEnter Choice: ";
cin >> choice;

switch (choice) {
    case 1: DisplayUppercaseASCII(); break;
    case 2: DisplayLowercaseASCII(); break;
    case 3: cout << "Exiting program...\n"; break;
    default: cout << "ERROR: Invalid Input - Enter a digit (1-3)...\n";
}
}

return 0;
}

// 2. Display uppercase letters (A-Z) with ASCII values
void DisplayUppercaseASCII() {
    for (char c = 'A'; c <= 'Z'; c++) {
        cout << c << " = " << int(c) << endl;
    }
}

// 3. Display lowercase letters (a-z) with ASCII values
void DisplayLowercaseASCII() {
    for (char c = 'a'; c <= 'z'; c++) {
        cout << c << " = " << int(c) << endl;
    }
}
```